

1. $E[X|Y] = E[X]$ with probability 1.

a) Show X and Y are uncorrelated.

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$\begin{aligned} E[XY] &= E[E[XY|Y]] = E[Y E[X|Y]] \\ &= E[Y E[X]] = E[X]E[Y] \end{aligned}$$

$$\therefore \text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 \quad ; \text{ uncorrelated.}$$

b) Give example that X and Y need not to be independent

$$Y = X^2, \quad X \sim \text{Unif}(-1, 1) \quad ; \quad X \text{ and } Y \text{ are dependent}$$

$$\text{Cov}(X, Y) = \text{Cov}(X, X^2) = E[X^3] - E[X]E[X^2] = 0. \quad \text{wikipedia covariance}$$

2. $X, Y \sim \text{Geo.}(p)$ independent

$$Z = E[X|X+Y]$$

$\therefore X, Y$ from same distribution

$$Z = E[X|X+Y] = E[Y|X+Y]$$

$$\therefore 2Z = E[X|X+Y] + E[Y|X+Y] = E[X+Y|X+Y] = X+Y.$$

$$Z = \frac{X+Y}{2}$$

$$E(Z) = \frac{1}{2}E[X+Y] = \frac{1}{2}(E[X] + E[Y]) = \frac{1}{2}\left(\frac{1}{p} + \frac{1}{p}\right) = \frac{1}{p}$$

$$\text{Var}(Z) = \text{Var}\left(\frac{X+Y}{2}\right) = \frac{1}{4}\text{Var}(X+Y) = \frac{1}{4}(\text{Var}(X) + \text{Var}(Y)) = \frac{1}{4}\left(2 \cdot \frac{1-p}{p^2}\right) = \frac{1-p}{2p^2}$$

3. X : number of fruits for first visit
 Y : number of fruits for second visit

$$P(X=1)=P(X=2)=P(X=3)=\frac{1}{3}, \quad P(Y|X=3)=0 \quad \forall y>0.$$

$$\begin{aligned} \text{a) } P(X+Y=3) &= P(X+Y=3|X=1) \times P(X=1) + P(X+Y=3|X=2) \times P(X=2) + P(X+Y=3|X=3) \times P(X=3) \\ &= P(Y=2|X=1) \times P(X=1) + P(Y=1|X=2) \times P(X=2) + P(Y=0|X=3) \times P(X=3) \\ &= \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{5}{9} \end{aligned}$$

$$\begin{aligned} \text{b) } P(Y>0 | X+Y=3) &= P(Y=1 | X+Y=3) + P(Y=2 | X+Y=3) \\ &= \frac{\frac{1}{9}}{\frac{5}{9}} + \frac{\frac{1}{9}}{\frac{5}{9}} = \frac{1}{5} + \frac{1}{5} = \frac{2}{5} \end{aligned}$$

$$\text{c) } E[K] \quad E[K|A]$$

K : total number of fruits.

$$A := \{K > 3\}$$

$$E[K] = E[X+Y] = \sum_{x,y} (x+y) P_{X,Y}(x,y)$$

$$= 2 \cdot P(1,1) + 3 P(1,2) + 3 P(2,1) + 4 P(1,3) + 5 P(2,3) + 4 P(2,2) + 3 \cdot P(3,0)$$

$$= 2 \cdot \frac{1}{9} + 3 \cdot \frac{1}{9} + 3 \cdot \frac{1}{9} + 4 \cdot \frac{1}{9} + 5 \cdot \frac{1}{9} + 4 \cdot \frac{1}{9} + 3 \cdot \frac{1}{3}$$

$$= \frac{20}{9} = 10/3$$

$$E[K|A] = \frac{4P(K=4) + 5P(K=5)}{P(K=4) + P(K=5)}$$

$$P(K=4) = P(X_1=1, X_2=3) + P(X_1=2, X_2=3) = 2/9$$

$$P(K=5) = P(X_1=2, X_2=3) = 1/9$$

$$\therefore E[K|A] = \frac{8/9 + 5/9}{3/9} = 13/3$$

$$d) \sigma_{K|A} = \sqrt{E[K^2|A] - E[K|A]^2}$$

$$E[K|A] = \frac{4 \cdot P(K=4) + 5 \cdot P(K=5)}{P(K=4) + P(K=5)} = 13/3$$

$$E[K^2|A] = \frac{16 \cdot P(K=4) + 25 \cdot P(K=5)}{P(K=4) + P(K=5)} = \frac{32/9 + 25/9}{3/9} = \frac{57}{3} = 19$$

$$\therefore \text{Var}(K|A) = 19 - (13/3)^2 = \frac{171 - 169}{9} = 2/9$$

$$\sigma_{K|A} = \sqrt{2/9} = \frac{\sqrt{2}}{3}$$

e) probability that she gets more than three fruits on each of next 16 days.

$$P(K \geq 3) = P(K=3) + P(K=4) + P(K=5) = 1 - P(K=2) = 8/9 = P(X)$$

$$P(X_1, X_2, \dots, X_{16}) = \prod_{i=1}^{16} 8/9 = (8/9)^{16} \approx 0.1519$$

$$f) X_i = K_i|A$$

$$E[X_1 + X_2 + \dots + X_{16}] = 16 E[X_i] = 16 \cdot \frac{13}{3} = \frac{208}{3}$$

$$4. \text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

$$\text{Var}(X) = E[X^2] - E[X]^2$$

$$= E[E[X^2|Y]] - E[E[X|Y]^2] + E[E[X|Y]^2] - E[E[X|Y]]^2$$

$$= E[E[X^2|Y] - E[X|Y]^2] + E[E[X|Y]^2] - E[E[X|Y]]^2$$

$$= E[\text{Var}(X|Y)] = \text{Var}(E[X|Y])$$

$$\text{show. } \text{Var}\left(\sum_{i=1}^N X_i\right) = E[N] \text{Var}(X_i) + E[X_i]^2 \text{Var}(N)$$

$$= E[\text{Var}\left(\sum_{i=1}^N X_i | N\right)] + \text{Var}\left(E\left[\sum_{i=1}^N X_i | N\right]\right)$$

$$= E[N \cdot \text{Var}(X_i)] + \text{Var}(N E(X_i)) = E[N] \text{Var}(X_i) + E[X_i]^2 \text{Var}(N)$$

5. Dobrow 5.22.

total population. 20,000
voters 12,000 $\left\{ \begin{array}{l} \text{democrats } 6,000 \\ \text{republicans } 5,000. \end{array} \right.$

200 exit poll
L voters. X : democrats

$$1) X \sim \text{Bin}\left(200, \frac{6,000}{12,000}\right)$$

$$E[X] = 200 \times \frac{1}{2} = 100 \quad \text{Var}(X) = 200 \times \frac{1}{2} \times \frac{1}{2} = 50$$

$$\text{Std}(X) = \sqrt{50} = 5\sqrt{2}$$

$$2) Y \sim \text{Bin}\left(200, \frac{5,000}{12,000}\right) \quad P_Y(y) = \binom{200}{y} \left(\frac{5}{12}\right)^y \left(\frac{7}{12}\right)^{200-y}$$

$$P(Y \geq 100)$$